Feature Engineering

Derived feature Importance

Signal extraction:

- Convert noisy raw data into measurable concepts (trend, volatility, regime, momentum, seasonality

Comparability and scaling:

- Standardize across units and magnitudes (z-scores, ranks, log transforms, - support comparison

Temporal structure:

- Characterise history (lags, rolling windows, exponentially weighted metrics

Robustness:

- Reduce sensitivity to outliers and non-stationarity

Model efficiency:

- Improve learnability for ML models by providing analytics-ready predictors that align with potential system dynamics

Derived features - Every Day Practical Taxonomy (domain specific)

Finance:

Returns & transforms: percent/log returns, k-step returns, diffs, ranks, z-scores

Trend & momentum: moving averages, MACD/MA oscillators, momentum, RSI/CMO/TRIX

Volatility & risk: TR/ATR, rolling/EWMA volatility, drawdown, tail proxies

Regime & structure: trend-strength (ADX/DMI), choppiness/VHF, change points, turning points

Liquidity/flow (needs volume): CMF/MFI/AD, volume oscillators, ROCV, pressure metrics

Relative value / spreads: cross-asset returns/spreads, sector dispersion, correlation/beta features

Macroeconomics:

Growth & inflation dynamics: QoQ/YoY growth, log-diffs, real vs nominal transforms

Trend–cycle decomposition: filtered trend/cycle (one-sided for causal), gap measures

Standardization & anomalies: rolling z-scores, percentile ranks, deviations from long-run mean

Lead–lag / composite indicators: lag stacks, diffusion indices, PCA/factors, financial conditions indices

Regime & breaks: recession indicators, change-point features, Markov switching proxies

Derived features - Keys to Derived Feature Classification

Frequency Keys

- **Daily+** - daily frequency and higher
- **Macro** - low-frequency macro series e.g. QoQ, YoY etc.
- **Not macro** = OHLCV related

Data Requirement Keys

- **OHLC:** (Open/High/Low/Close)
- **OHLCV:** (OHLC + Volume)
- **Close-only:** (single value series per period)
- **Volume-only:** (volume series per period)

Core Base Transforms

Feature name	Short description	Domain	Analytic
Difference	Absolute change versus prior period	F+M	np+pd
k-step Difference	Absolute change versus k periods ago	F+M	np+pd
Percent change (1-step)	Relative change versus prior period	F+M	np+pd
k-step Percent change	Relative change versus k periods ago	F+M	np+pd
Log transform	Stabilizes scale for strictly positive series	F+M	np+pd
Log return (1-step)	Relative change on a log scale versus prior period	F+M	np+pd
k-step Log return	Relative change on a log scale versus k periods ago	F+M	np+pd
Rolling mean	Local average over a trailing window	F+M	pd
Rolling standard deviation	Local variability over a trailing window	F+M	pd
Rolling z-score	Standardized deviation from a trailing-window mean	F+M	pd
EWMA / EMA	Exponentially weighted moving average (more weight on recent data)	F+M	pd
Rolling min	Lowest value in a trailing window	F+M	pd
Rolling max	Highest value in a trailing window	F+M	pd
Rolling range	Window spread (max minus min) as a dispersion proxy	F+M	pd

Legend: F – Financial Markets, **M** – Macroeconomics, **pd** – Pandas, **np** – Numpy.

Advanced Transforms - 1

Volatility & Range			
Feature name	**Short description**	**Domain**	**Analytic**
True Range	Intraperiod range measure that also captures gap effects	F	pd
Average True Range (ATR)	Smoothed true range; standard volatility proxy	F	Pd
Trend Strength and Direction			
Directional Movement Index (DMI / DX)	Directional movement pressure used to characterize trending behavior	F	pd
Average Directional Index (ADX)	Trend strength metric derived from directional movement	F	pd
Aroon Oscillator	Trend timing based on how recently highs and lows occurred	F	pd
Choppiness Index	Identifies trending versus range-bound conditions	F	pd
Vertical Horizontal Filter (VHF)	Trend versus noise ratio; trend strength proxy	F+M	Pd
Momentum	Direction and magnitude of change over a chosen lookback	F+M	np+pd
Moving Average Oscillator	Difference between fast and slow moving averages	F+M	pd
MACD	Momentum/trend indicator based on moving average convergence/divergence	F+M	pd
Relative Strength Index (RSI)	Bounded momentum indicator based on average gains and losses	F+M	pd
Stochastic Oscillator	Position of close within a recent high–low range	F	pd
Slow Stochastic Oscillator	Smoothed stochastic measure to reduce noise	F	pd
Stochastic RSI	Stochastic-style normalization applied to RSI values	F+M	pd
Commodity Channel Index (CCI)	Deviation-from-average oscillator often used for cyclical behavior	F	pd
Chande Momentum Oscillator (CMO)	Momentum oscillator related to the balance of up/down moves	F+M	pd
TRIX	Smoothed momentum indicator designed to reduce short-term noise	F+M	pd
Detrended Price Oscillator (DPO)	Emphasizes shorter cycles by removing longer trend components	F+M	pd
Williams %R	Overbought/oversold style oscillator similar to stochastic variants	F	pd

Advanced Transforms - 2

Volume, Money Flow, and Accumulation			
Feature name	**Short description**	**Domain**	**Analytic**
Accumulation Distribution (AD)	Volume-weighted accumulation proxy based on close location in range	F	pd
Chaikin Money Flow (CMF)	Money flow pressure over a window using price location and volume	F	pd
Chaikin Oscillator	Oscillator derived from accumulation/distribution dynamics	F	pd
Money Flow Index (MFI)	Volume-weighted oscillator similar in spirit to RSI	F	pd
Ease of Movement (EMV)	Volume-adjusted “ease” of price movement	F	pd
Force Index (FI)	Price change scaled by volume to measure pressure	F	pd
Volume Oscillator	Fast–slow moving average comparison of volume	F	pd
Rate of Change Volume (ROCV)	Momentum of volume over a lookback window	F	Pd
Volatility and Range Variants			
Volatility Ratio	Ratio-style volatility measure; definition varies by library	F+M	pd
Volatility Indicator (VI)	Often refers to vortex-style measures; definition varies	F	pd

Advanced Transforms - 2

feature name	python function (no parameters)
Difference	pd.Series.diff
k-step Difference	pd.Series.diff
Percent change (1-step)	pd.Series.pct_change
k-step Percent change	pd.Series.pct_change
Log transform	np.log
Log return (1-step)	custom.log_return
k-step Log return	custom.log_return
Rolling mean	pd.Series.rolling
Rolling standard deviation	pd.Series.rolling
Rolling z-score	custom.rolling_zscore
EWMA / EMA	pd.Series.ewm
Rolling min	pd.Series.rolling
Rolling max	pd.Series.rolling
Rolling range	custom.rolling_range
True Range	ta.true_range
Average True Range (ATR)	ta.atr
Directional Movement Index (DMI / DX)	ta.adx
Average Directional Index (ADX)	ta.adx
Aroon Oscillator	ta.aroon
Choppiness Index	ta.chop
Vertical Horizontal Filter (VHF)	ta.vhf
Momentum	ta.mom
Moving Average Oscillator	ta.ppo
MACD	ta.macd

feature name	python function (no parameters)
Relative Strength Index (RSI)	ta.rsi
Stochastic Oscillator	ta.stoch
Slow Stochastic Oscillator	ta.stoch
Stochastic RSI	ta.stochrsi
Commodity Channel Index (CCI)	ta.cci
Chande Momentum Oscillator (CMO)	ta.cmo
TRIX	ta.trix
Detrended Price Oscillator (DPO)	ta.dpo
Williams %R	ta.willr
Accumulation Distribution (AD)	ta.ad
Chaikin Money Flow (CMF)	ta.cmf
Chaikin Oscillator	ta.adosc
Money Flow Index (MFI)	ta.mfi
Ease of Movement (EMV)	ta.eom
Force Index (FI)	ta.efi
Volume Oscillator	ta.pvo
Rate of Change Volume (ROCV)	ta.roc
Chaikin Volatility	ta.cvi
Volatility Ratio	custom.volatility_ratio

Functions access: **import pandas as pd**

! import pandas_ta_classic as ta

To Summarise Feature Engineering

Tier 1: Core Base Transforms

- Differences, percent/log returns, log transforms, rolling statistics (mean, std, min/max/range), rolling z-scores, EWMA/EMA and other one-sided smoothers
- **Purpose:** normalize scale, encode history, stabilize non-stationarity, provide multi-scale structure

Tier 2: Domain indicators

- Finance: RSI, MACD, ATR/TR, ADX/DMI, stochastic family, flow/volume metrics (CMF/MFI/AD), channels/regime proxies
- Macro: growth rates, trend–cycle decomposition, diffusion/factor indices, regime/break features
- Run transforms across **frequency-dependent window grids** (e.g., daily 5/10/20/63/126/252; monthly 3/6/12/24; quarterly 4/8/12)
- Multi-horizon features capture **short / medium / long** dynamics and reduce reliance on any single lookback
- **Purpose:** map domain concepts, improves interpretability, helps models to learn faster

Considerations:

- Feature naming, meta data, especially for persistent features e.g. **US Implied Vol SPX 3m50dc_diff_5**, **y_win_20_dd_10**
- Leakage – Awareness of centred/two-sided smoothing and any forward-looking computation (right labelling)
- Feature explosion and consequential collinearity
- Regime dynamic dependence - trending vs mean-reverting regimes
- Underlying analytics software version control
- Core input data revisions (macroeconomics)